



MOONSHOT

44

REGULATOR AI GLOBAL • AUGUST 2026

Don Quixote's Windmills

Building Second-Generation AI for the Bounded Enterprise

Probabilistic intelligence is valuable because it can imagine. It becomes dangerous when imagination acquires authority.

A narrative white paper on the deterministic discipline required for consequential artificial intelligence

AI Blue Box • Capability Airlock

Zachary Michael Akins

Founder and Chief Executive Officer

Regulator AI Global, Inc.

Bruce Perens

Strategic Advisor, Regulator AI Global

Co-founder, Open Source Initiative

THE SECOND GENERATION

The fluid's run out of my brakes

A terrifying statement indeed—the prelude to a rapid, out-of-control trip, perhaps a messy death. This is the exact situation of AI-mediated business today, caused by the fundamental features of first-generation artificial intelligence.

A first-generation AI does not reliably know the difference between instructions and data. It receives input and can mix the two together. Perpetrators and advertisers exploit this fault, whispering their own commands to the machine that increasingly runs your business. Is it acting in your best interest, or theirs?

Artificial intelligence, of any generation, is probabilistic. It is astonishing that it works as well as it does, giving the impression of intelligence through a vast and repeated form of roulette. You would not trust your company's future to a casino table, but this is what first-generation AI effectively asks you to do: probability means you win and lose. It produces wrong answers as well as right ones, sincere and credible explanations that frame mistakes as achievements, and “facts” pulled from thin air and expressed with great confidence.

Artificial intelligence with discipline

Regulator AI has the answer: second-generation AI is artificial intelligence with discipline. It reliably separates critical system instructions that maintain integrity from user commands, and separates outside data from either, so that a treacherous whisper from outside is never treated as an instruction. It adds circuit breakers that stop bad data, wrong assumptions, and hallucinations from risking a company—or worse.

Regulator AI has developed many inventions that advance artificial intelligence into this second generation: 60 patent applications in process as this paper is written, probably a dozen more by the time it is read. Each is deeply complicated, but their combined effect is like adding a bridle and harness, saddle and stirrups to a horse, turning a flighty beast into a reliable worker. Or like building a modern electrical code onto the deadly early system of knife switches and open wiring.

Each invention addresses a different source of current unreliability, adding control that makes artificial intelligence suitable for financial trading, military planning, space guidance, and surgery. Together, the inventions mesh into a coherent system of governance: recording inputs and conclusions for later audit; limiting excursions, whether space maneuvers or financial trades, to sensible boundaries; bringing humans into the approval loop when appropriate; and ensuring that a system showing oddities has no path to control.

One example: the Material Safety Layer

The Material Safety Layer is one of those pieces. Artificial intelligence no longer gives advice alone. Consider an AI tasked with changing the orbit of one satellite in a constellation of 10,000. It may have to act because a collision could contribute to a Kessler collapse, making space inaccessible for decades. It cannot necessarily wait for a human. Yet it also must not make an out-of-bounds decision carrying risks comparable to inaction.

The Material Safety Layer is not itself an AI. It is a much simpler system that can be trusted to act reliably where probabilistic artificial intelligence, on its own, cannot. Their combination is one element of second-generation AI. The layer constrains action to reasonable boundaries and can raise an alarm,

transfer the task to another system, or call for human assistance when those boundaries cannot be respected.

Now extend that form of governance across the AI's activity: layers that preserve information in durable, uneditable form so an audit can reconstruct what happened; layers that detect patterns of failure; layers that ensure actions are reasonable and do not create unaccounted-for risks to a company—or to humanity.

Such governance does more than make artificial intelligence reliable. By constraining risk to known boundaries and producing evidence that the boundaries held, it can make an AI-enabled company insurable and investable.

The bounded enterprise

The Material Safety Layer is only one example. Across the portfolio, the governing idea appears in different forms: probabilistic intelligence may propose, interpret, and act, but consequential authority must remain bounded by systems that can be independently trusted.

In this paper's governing metaphor, first-generation AI is Don Quixote with direct control of the horse and lance: a vivid interpretation can become consequential action. Second-generation AI preserves the imagination while placing authority, reachability, and proof under deterministic supervision. At the enterprise level, that reduces to four questions.

Four questions every AI-dependent business must answer

CONTROL QUESTION	ENTERPRISE MEANING
Influence	What information may shape the model's reasoning?
Authority	Which outputs are permitted to alter protected state or direct action?
Reachability	Which records, tools, funds, accounts, and external systems can the process physically reach?
Proof	What independent evidence establishes that every required condition was satisfied?

THREE BUSINESSES • ONE ARCHITECTURAL WEAKNESS

Sammy: when a helpful answer enters the books

Sammy provides financial administration and reconciliation services. Its AI system reads contracts, invoices, correspondence, payment records, and prior transactions. It is exceptionally good at finding patterns and proposing how mismatched records should be resolved.

One morning the model encounters an amended contract embedded in an email chain. It correctly summarizes the economic terms but treats explanatory language as an authorized instruction. The proposed adjustment flows into a downstream workflow. No employee intended to approve it. The system nevertheless had enough reach to make the recommendation operationally consequential.

The loss is larger than the transaction

Sammy can reverse the entry. It cannot so easily reverse the questions the incident creates. Were other entries produced the same way? Can the audit population be reconstructed? Did management know that the model could translate semantic relevance into operational influence? Was the control failure disclosed to the insurer? Are financial representations made to lenders or buyers still reliable?

What begins as a hallucination becomes an internal-control deficiency. That deficiency can increase audit cost, narrow insurance coverage, trigger contractual notice duties, delay financing, invite regulatory scrutiny, and reduce the confidence applied to Sammy's valuation. A buyer may discount not only the disputed transaction but every automated process whose authority cannot be proven.

PROBLEM A Sammy does not merely need a more accurate model. It needs a provable separation between what the model understands and what the enterprise authorizes.

Crossing the first threshold

Under the AI Blue Box architecture, semantic relevance does not create operational authority. Content may inform analysis without becoming an executable instruction. A model may recommend a journal treatment, explain its reasoning, and cite supporting material while remaining unable to change the authoritative ledger unless separate policy, identity, provenance, and permission conditions are satisfied.

The business consequence is not perfection. It is containment. A mistaken interpretation can remain visible for review without silently becoming a protected-state change. Sammy gains a defensible answer to the question that matters after an incident: not merely whether the model was wrong, but whether the architecture prevented that wrong answer from acquiring authority.

THE AUTHORITY BOUNDARY

River: when probability can reach the money

River operates live wagering markets. It ingests game events, pricing information, market behavior, account status, jurisdictional rules, and settlement conditions. Speed matters. Delay creates commercial loss. Yet every market action and settlement also depends on facts that must be authoritative, timely, and permitted in the relevant jurisdiction.

During a live event, a probabilistic process interprets an ambiguous data signal as final. Its explanation is reasonable. The signal later proves provisional. If that interpretation can reach market controls or settlement systems directly, the error may propagate across thousands of accounts before an operator can intervene. Customers may be paid incorrectly, markets may be suspended late, and inconsistent treatments may appear across jurisdictions.

Reachability changes the risk

A bad internal answer is a quality problem. A bad answer with access to pricing, settlement, customer communication, or funds is an operational event. River's risk is therefore determined not only by model accuracy but by the path between model output and protected action.

PROBLEM B The relevant control is not “Did the model sound confident?” It is “Was there an authorized path from this output to that action?”

The AI Blue Box architecture treats authority and reachability as distinct from semantic fluency. The AI may analyze every relevant signal while remaining structurally unable to alter protected state through an unauthorized path. Operational influence can be forced to zero when authority is absent or the destination is outside the permitted reachability graph.

What River can tell its stakeholders

- Regulators: interpretations cannot silently become settlement authority.
- Insurers: covered workflows have explicit control boundaries and reconstructable decision provenance.
- Customers: ambiguous events can be held, reviewed, or reversed before irreversible release.
- Investors and buyers: exposure is bounded by architecture rather than by a promise that the model will usually behave.

River still uses probabilistic intelligence to move quickly. The difference is that speed no longer depends on giving an interpretive system unrestricted access to the machinery of the business.

THE GUARANTEE BOUNDARY

Scarlett: when “complete” is not complete

Scarlett uses agentic AI throughout the enterprise. Agents review diligence materials, coordinate vendors, prepare regulatory submissions, update customer records, and execute multi-step workflows. Management sees dramatic productivity gains. It also begins to rely on the agents’ declarations that work is finished.

Scarlett then acquires a smaller company. During diligence, an AI agent is instructed to examine every material customer agreement and identify change-of-control provisions. The agent reviews most of the repository, generates an excellent summary, and reports completion. Several agreements stored under an unexpected naming convention were never processed. One contains a termination right that materially changes the economics of the transaction.

The model did not necessarily fabricate a clause. It committed a different form of hallucination: it represented a structurally incomplete task as complete. Because the output was articulate and the omissions were invisible, the declaration passed as proof.

PROBLEM C A model that performed the work cannot be the sole authority certifying that every required item was processed.

Crossing the second threshold

The Capability Airlock begins with a simple rule: the AI that performed the work cannot be the sole authority declaring it complete. When a task demands certainty rather than a plausible answer, the result remains inside the airlock until something independent confirms it.

For Scarlett, the practical question is straightforward: did the system examine every agreement it was assigned to examine? The AI may find, interpret, and summarize the contractual risk. Independent verification determines whether the work is ready to leave the airlock.

ONE HYBRID OPERATING SYSTEM

From model governance to compliance by construction

Sammy, River, and Scarlett appear to have different problems. Sammy confused relevant language with authorized instruction. River allowed interpretation to approach protected execution. Scarlett treated persuasive work product as proof of exhaustive completion. All three failures arise because a probabilistic system was asked to provide something probability alone cannot guarantee.

AI Blue Box and Capability Airlock form an interlocking defensive and offensive framework. AI Blue Box governs the boundary between meaning and authority. Capability Airlock governs the boundary between probabilistic work and structurally guaranteed completion. One limits what an interpretation can cause. The other determines what must be proven before a result can be certified or released.

LAYER	WHAT IT CONTRIBUTES	WHAT IT CANNOT DECLARE
Probabilistic intelligence	Interpretation, pattern recognition, synthesis, recommendation	Authority, exhaustive completion, or structural correctness
AI Blue Box	Policy-controlled influence, authority, reachability, and protected-state separation	That semantic relevance itself authorizes action
Capability Airlock	Independent verification when work must be exact or complete	That an AI-generated result is ready for release

Defensive and offensive at the same time

Defensively, the framework contains unauthorized action, false completion, irreproducible state changes, and uncontrolled propagation. Offensively, it allows an enterprise to deploy AI into higher-value work with greater confidence. The same boundary that prevents a model from releasing an unsupported transaction permits the model to analyze more data, propose more options, and operate closer to the business without requiring blind trust.

STRATEGIC RESULT The objective is not less AI. It is more useful AI operating inside boundaries that remain real under pressure.

CAPITAL CONSEQUENCES

Insurance, valuation, financing, and acquisition

AI governance is often described as an ethics, compliance, or information-technology concern. That framing is already too narrow. As AI becomes embedded in revenue, accounting, contracting, underwriting, settlement, safety, and customer operations, the quality of the control architecture becomes a capital-allocation issue.

Insurance

Insurers cannot price an exposure they cannot bound. If a model's practical authority depends on informal prompts, changing integrations, discretionary human review, or undocumented tool access, the carrier cannot readily determine the frequency or severity of loss. Explicit authority boundaries, deterministic release conditions, append-only evidence, and reconstructable state transitions do not eliminate claims. They create facts from which coverage, exclusions, limits, deductibles, and controls can be evaluated.

Enterprise valuation and financing

A company that derives earnings from uncontrolled agentic processes may be carrying hidden operational leverage. Reported efficiency can disappear after one material incident or after a lender requires manual controls. A bounded system makes the source and limits of automation more legible. That can reduce uncertainty in forecasts, improve the credibility of representations, and distinguish durable operating advantage from temporary model performance.

Mergers and acquisitions

Traditional diligence asks what software a target uses, what data it holds, and whether material controls exist. AI-dependent diligence must also ask what outputs can acquire authority, which protected systems are reachable, which tasks require structural guarantees, and whether completion evidence can be independently reconstructed. Without those answers, a buyer may inherit latent model risk that was never reflected in price, escrow, indemnity, insurance, or integration planning.

Governance and personal accountability

Boards and executives increasingly face a simple evidentiary question after an AI-related loss: What prevented this system from doing what it did? Policies, committees, and training may show attention. They do not by themselves demonstrate that the system was technically unable to cross a prohibited boundary. Compliance by construction supplies a stronger answer because the relevant conditions are enforced within the operational architecture.

OPERATIONAL IMPLEMENTATION

The architecture is ready for deployment

The bounded enterprise does not require organizations to discard their existing models, databases, identity systems, workflow platforms, or security infrastructure. The architecture introduces a supervisory operating layer at the exact points where probabilistic intelligence approaches operational authority, protected-state change, or a result requiring structural proof.

1. Define	Identify protected systems, authoritative records, operational domains, and the decisions for which probabilistic output may inform—but may not independently authorize—action.
2. Separate	Establish explicit boundaries among semantic influence, operational authority, reachability, and protected-state change. Remove authority that exists only because a model can reach a tool.
3. Route	Direct tasks requiring exact coverage, ordering, arithmetic, dependency enforcement, persistent state, or completeness proof into deterministic capabilities designed to supply those guarantees.
4. Certify	Require independent validation and committed evidence before a result can be represented as complete, authoritative, or releasable.
5. Scale	Extend governed authority across additional workflows and operating domains while preserving the same enforceable boundaries and proof requirements.

The next phase is implementation

The underlying problem is no longer theoretical, and neither is the architecture required to address it. The work ahead is operational: placing enforceable boundaries into enterprise systems that are already moving from AI assistance toward AI authority.

A relevant document cannot become an authorized instruction merely because a model interpreted it that way. An available tool cannot become a permitted action merely because an agent can reach it. An incomplete task cannot become complete merely because the model says the work is finished. A protected-

state change cannot be released without the authority and evidence the enterprise predetermined it would require.

Over the coming months, these architectures can move into operating environments as the control layer between probabilistic intelligence and consequential action. The transition does not depend on eliminating hallucinations. It depends on ensuring that hallucinations cannot acquire authority, cross protected boundaries, or substitute narrative confidence for proof.

The precise invariants, token structures, gate sequences, state machines, recovery logic, and enforcement configurations remain proprietary. What is public—and now unavoidable—is the governing principle: probability may generate the possibility, but authority and completion must be established elsewhere.

CONCLUSION

The bounded enterprise

First-generation AI has been asked to become more capable, more connected, and more autonomous. Each advance increases value. Each also increases the consequence of confusing a probable interpretation with an authorized fact, an available tool with a permitted action, or a fluent report with proof of completion.

Sammy's exposure begins when explanatory language enters the books as authority. River's begins when an ambiguous interpretation can reach markets, settlements, or money. Scarlett's begins when an agent can omit part of a defined population and still declare the task complete. Better models may reduce the frequency of these events. They do not change their architectural character.

BOARD-LEVEL QUESTION **The decisive question is no longer whether AI will sometimes hallucinate. It is whether a hallucination can acquire authority, reach protected systems, or certify itself as complete.**

AI Blue Box addresses the first boundary: semantic intelligence may inform the enterprise without automatically receiving operational authority or unrestricted reachability. Capability Airlock addresses the second: work requiring structural guarantees is routed to deterministic capabilities, and completion is independently validated before certification and release.

The mathematics stands on established disciplines: capability security, graph reachability, deterministic computing, state-machine control, set-based completeness, and append-only evidence.

The result is the operating model for second-generation enterprise AI. Probability handles ambiguity, language, pattern, and possibility. Deterministic supervision handles authority, reachability, invariants, protected-state transition, and proof. Neither replaces the other. Each performs the work it is structurally suited to perform.

For insurers, the framework makes exposure more legible. For investors and lenders, it distinguishes durable automation from uncontrolled leverage. For buyers, it turns AI diligence into an examination of enforceable boundaries. For regulators and boards, it creates evidence that policy survived contact with operations. For management, it opens higher-value uses without requiring faith that the model will never be wrong.

Don Quixote may continue to see giants. The bounded enterprise simply ensures that the story cannot move the horse, wield the lance, or declare victory without independent authority and proof.